



BEN LUBETZKY

MACHINE LEARNING / FRONTEND / BACKEND

— PROFILE

Computer science student in Porto. Through personal projects I work with computer vision models, LLM-based systems, and APIs with the web interfaces on top of them, building from research papers and reference implementations.

— SELECTED PERSONAL PROJECTS

LedgerDrop

Invoice-processing web app: async upload pipeline on FastAPI and SQLAlchemy, a Next.js frontend, and a designed path through extraction, validation and human review.

Python FastAPI PostgreSQL SQLAlchemy Alembic Next.js TypeScript Docker

Customer Support Agent

Agentic retail-support assistant with policy RAG, tool calling, identity checks, guarded order actions and human escalation.

Python FastAPI RAG Tool calling ChromaDB SQLite

Research Paper Assistant

Local RAG pipeline over computer-vision papers: structure-aware chunking, hybrid retrieval, metadata reranking and grounded, cited answers.

Python FastAPI Hybrid retrieval Embeddings ChromaDB PDF chunking

ArcFace Face Recognition

Closed-set face identification on LFW with a pretrained ResNet-50 backbone and additive angular-margin loss.

Python PyTorch Torchvision ResNet-50 Angular-margin loss scikit-learn

— TECH STACK

Languages

Python, TypeScript, JavaScript, SQL

ML, vision & LLMs

PyTorch, Torchvision, scikit-learn, NumPy, CNNs, transfer learning, embeddings, OpenAI API, RAG, ChromaDB

Backend, frontend & tooling

FastAPI, REST, SQLAlchemy, PostgreSQL, Next.js, React, Docker, Git, pytest

— EDUCATION

BSc Computer Science

University of Porto (FCUP)
2024–present, third year

BSc Computer Science coursework

Open University of Israel
2023–2024

— LANGUAGES

Hebrew — Native

English — Native

Portuguese — C1

French — B1

Spanish — B1